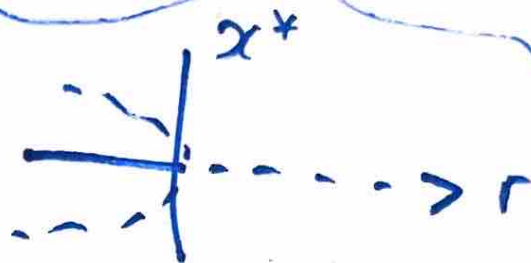


$$\dot{x}(t) = x(r + x^2 - x^4)$$

$$\dot{x} = x(r + x^2)$$



$$\dot{u} = a u + b u^3 - c u^5$$

$$, a, b, c > 0$$

$$x_1^* = 0$$

$$-x^4 + x^2 + r = 0$$

$$x^2 - x^2 - r = 0$$

$$x_{1,2}^2 = \frac{1 \pm \sqrt{1 + 4r}}{2}$$

2'

$$\dot{x} = f(x, r)$$

$$f(x, r) = rx + x^3 - x^5$$

$$x_1^* = 0$$

$$x_{2,3}^* = \pm \sqrt{\frac{1 - \sqrt{1 + 4r}}{2}}$$

$$r > -\frac{1}{4}$$

$$x_{4,5}^* = \pm \sqrt{\frac{1 + \sqrt{1 + 4r}}{2}}$$

$$\frac{\partial}{\partial x} f(x, r) = r + 3x^2 - 5x^4$$

$$x_1^* = 0: \left. \frac{\partial f}{\partial x} \right|_{x_1^* = 0} = r$$

unstable $r > 0$
stable $r < 0$

3

$$x_2 = \sqrt{\frac{1 - \sqrt{1+4r}}{2}}$$

$$\frac{\partial f}{\partial x} = r + 3x^2 - 5x^4 = 0 \Rightarrow r \geq -\frac{1}{4}$$

$$= r + \frac{3}{2} \left(1 - \sqrt{1+4r} \right) - \frac{5}{4} \left(1 - \sqrt{1+4r} \right)^2$$

$$= r + \frac{3}{2} - \frac{3}{2} \sqrt{1+4r} -$$

$$- \frac{5}{4} \left(1 + 1 + 4r - 2\sqrt{1+4r} \right)$$

$$= r + \frac{3}{2} - \frac{3}{2} \sqrt{1+4r} - \frac{5}{2} - 5r +$$

$$= -4r - 1 + \sqrt{1+4r} + \frac{5}{2} \sqrt{1+4r}$$

$$= -(1+4r) + \sqrt{1+4r}$$

$$\frac{-4 \pm \sqrt{1+4r}}{2} - (1+4r) = x$$

$$r > -\frac{1}{4}$$

$$\emptyset > r > -\frac{1}{4}$$

$$r > \emptyset$$

$$x > \emptyset$$

$$x < \emptyset$$

5

$$x_2 = \sqrt{\frac{1 - \sqrt{1 + 4r}}{2}}$$

$$\frac{\partial x_2}{\partial r} = \frac{-1}{\sqrt{2\sqrt{1 - \sqrt{1 + 4r}}}} \cdot \frac{1}{\sqrt{1 + 4r}}$$

$$\frac{\partial x_2}{\partial r} \Big|_{r=0} = \infty$$

$$\frac{\partial x_2}{\partial r} \Big|_{r=-1/4} = \infty$$

$$-6' \\ x_4 = \sqrt{\frac{1 + \sqrt{1+4r}}{2}}$$

$$\frac{\partial f}{\partial x} \big|_{x_4} =$$

$$= r + 3x^2 - 5x^4$$

$$= r + \frac{3}{2}(1 + \sqrt{1+4r}) - \frac{5}{4}(1 + \sqrt{1+4r})^2$$

$$= r + \frac{3}{2} + \frac{3}{2}\sqrt{1+4r} - \frac{5}{4}(1 + 1 + 4r + 2\sqrt{1+4r})$$

$$= r + \frac{3}{2} + \frac{3}{2}\sqrt{1+4r}$$

$$- \frac{5}{2} - 5r - \frac{5}{2}\sqrt{1+4r}$$

$$= -4r - 1 - \sqrt{1+4r} < 0 \quad r > -\frac{1}{4}$$

-7-

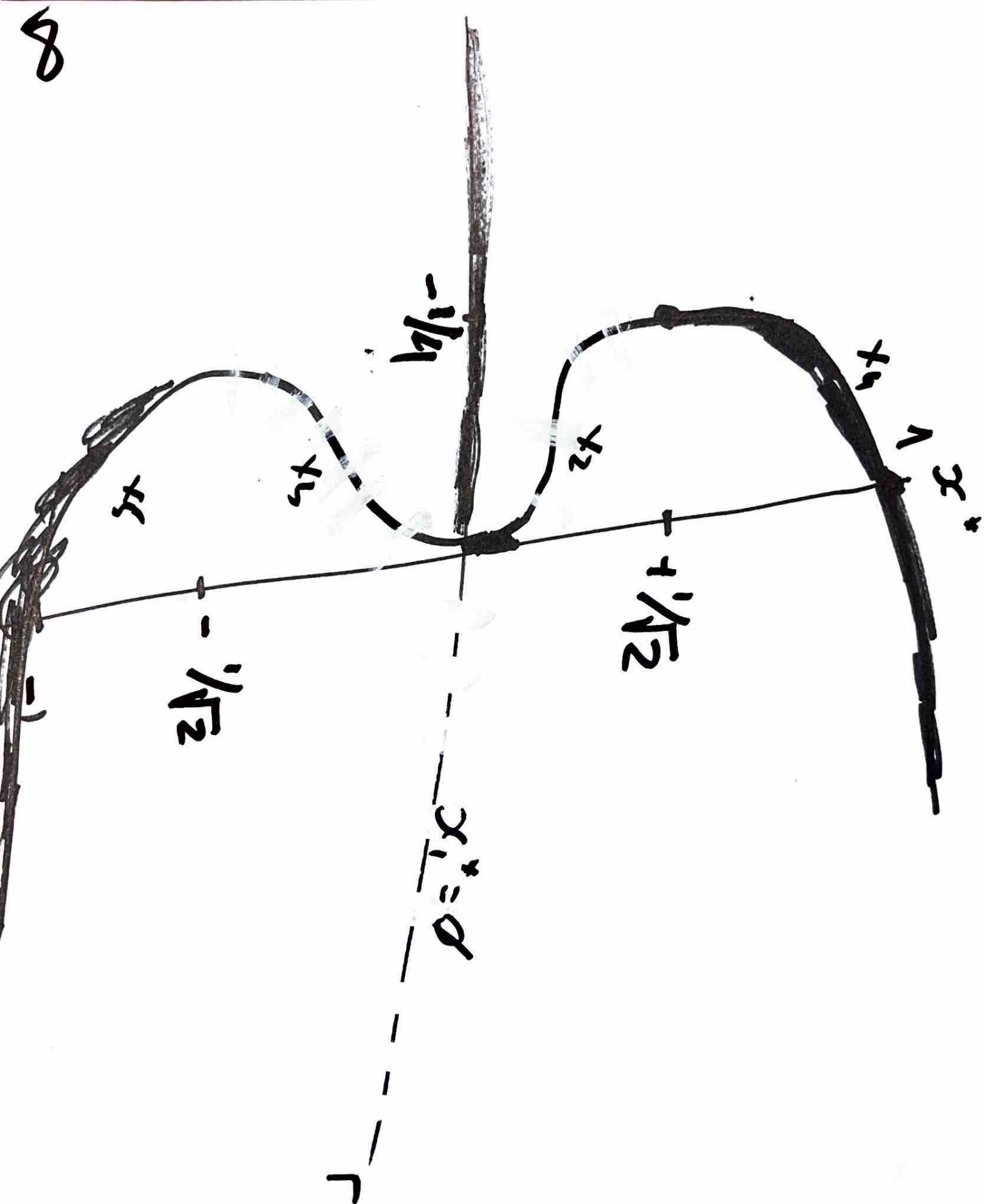
$$\frac{\partial x_4}{\partial r} = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{1 + \sqrt{1 + 4r}}} \frac{1}{\sqrt{1 + 4r}}$$

$$r = -\frac{1}{4}; \quad \left. \frac{\partial x_4}{\partial r} \right|_{r = -\frac{1}{4}} = \infty$$

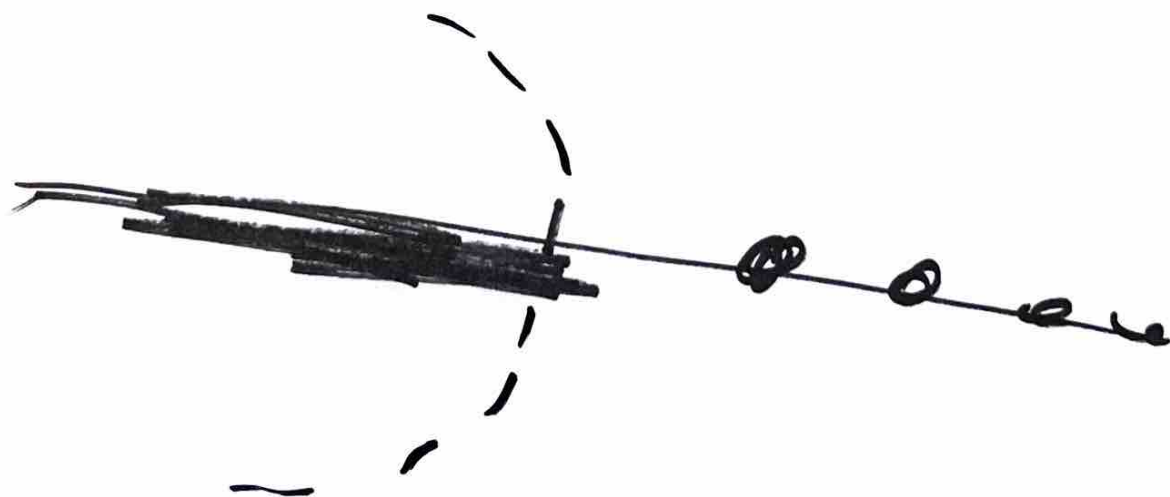
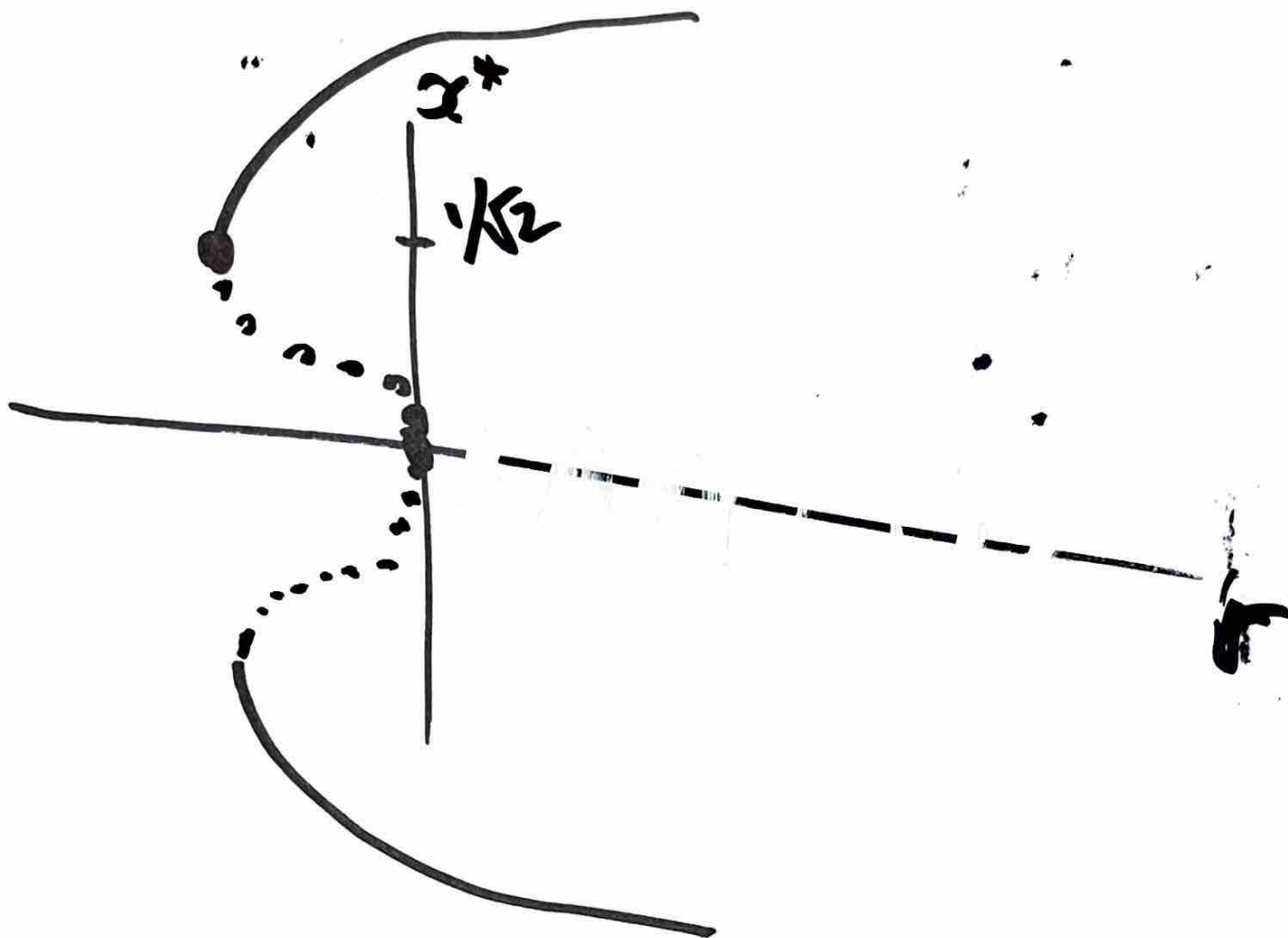
$$x_2(r = -\frac{1}{4}) = \frac{1}{\sqrt{2}}; \quad x_2(r = 0) = 0$$

$$x_4(r = -\frac{1}{4}) = \frac{1}{\sqrt{2}}$$

8



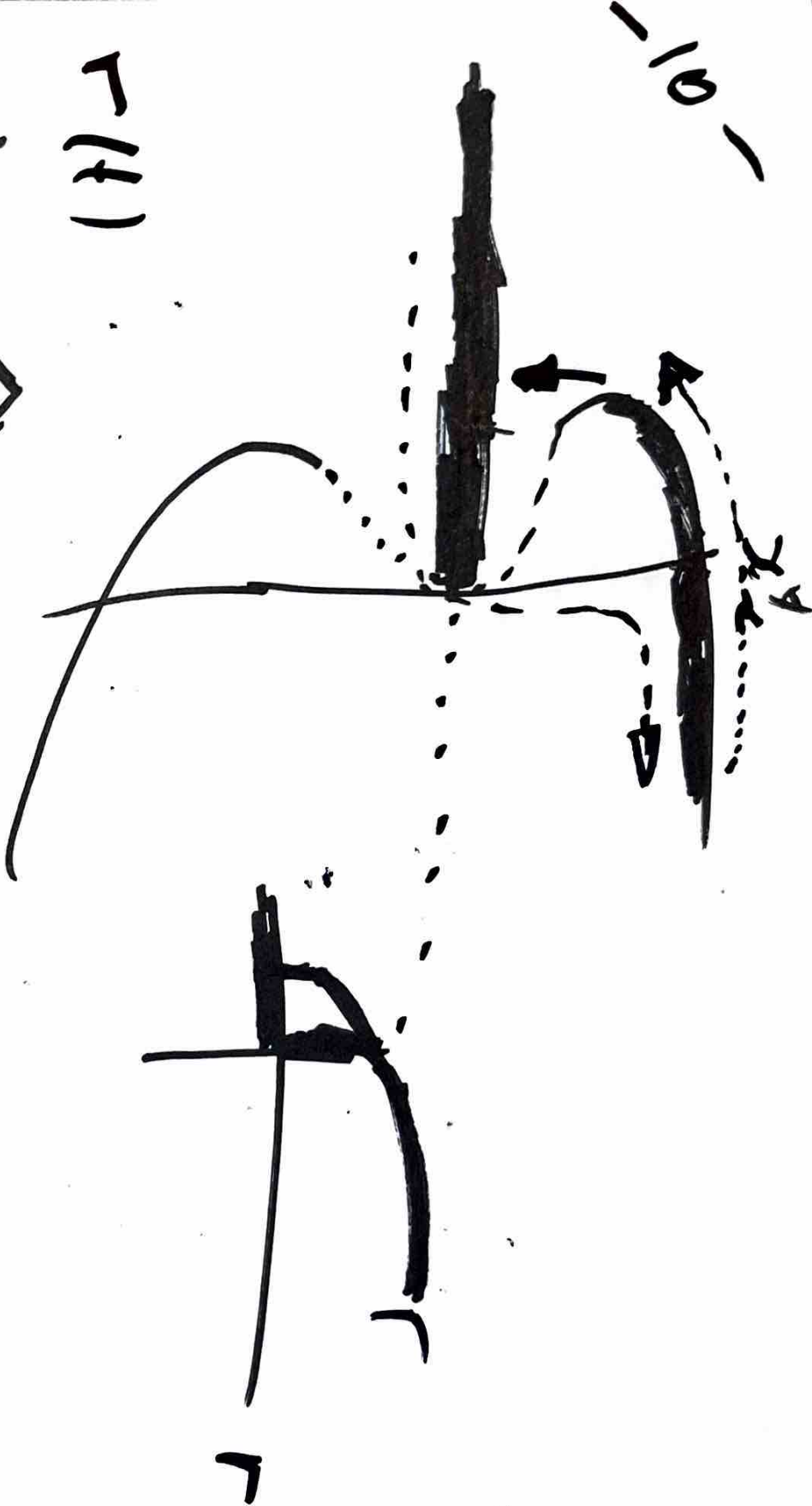
9

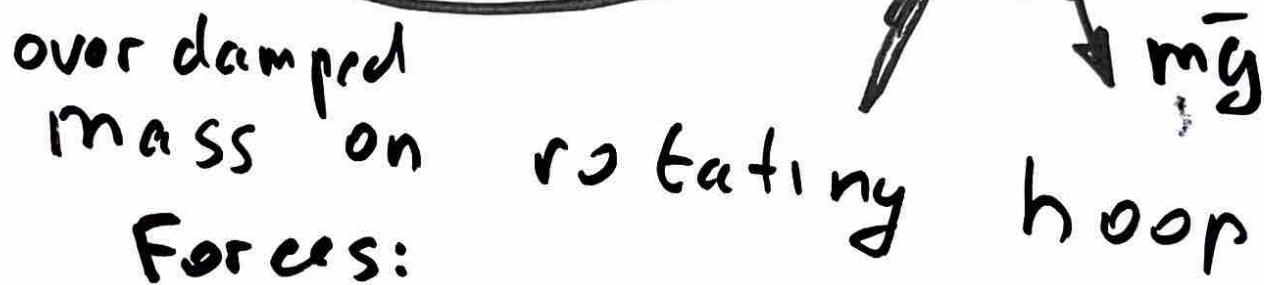


hysteresis



$t(t)$




$$\vec{F} = m\vec{g} + m\vec{\omega}^2 \mathbf{g} = m \cdot R \ddot{\psi}$$

$$m R \ddot{\psi} = \underbrace{-b \dot{\psi}}_{\text{friction}} - \underbrace{mg \sin \psi}_{+ R m \omega^2 \cos \psi \sin \psi}$$

$$F_c = m\omega^2 R$$

2-

$$m R \ddot{\varphi} = - \theta g + m g \sin \varphi + m \omega^2 R \sin \varphi \cos \varphi$$

Unconstrained a particle

$$m R \ddot{\varphi} = \theta$$

$$m \ddot{\varphi} = \theta$$

$$\theta g = - m g \sin \varphi + m \omega^2 R \sin \varphi \cos \varphi$$

$$\dot{\varphi} = \omega \sin \varphi - \left[\theta + \omega^2 R \cos \varphi \right]$$

$$\sin \varphi = \theta, \quad \varphi^* = \theta, \quad \varphi, \dots$$

$$-g + \omega^2 R \cos \varphi = \theta \quad \cos \varphi = g / \omega^2 R$$

$$\dot{\varphi} = \frac{m}{b} \sin \varphi (-g + \omega^2 R \cos \varphi)$$

$$\varphi_1 = \pi \cdot n; \quad n = 0, \pm 1, \pm 2, \pm 3, \dots$$

$$\varphi_2 = \text{Arc Cos} \left(\frac{g}{\omega^2 R} \right) + \pi n$$

$$\dot{\varphi} = f(\varphi, \omega) = \frac{m}{b} \sin \varphi (-g + \omega^2 R \cos \varphi)$$

$$\frac{\partial f(\varphi, \omega)}{\partial \varphi} = -\frac{mg}{b} \cos \varphi$$

$$+ \frac{m}{b} \cos^2 \varphi \cdot \omega^2 R - \frac{m}{b} \omega^2 R \sin^2 \varphi$$

13

$$\underline{F} = m \omega^2 r = m \omega^2 \bar{R} \sin \varphi$$

$$(x), B(x) f + (x) B(x), f =$$

$$\frac{\nabla}{1} [(x) B(x) f - (x), B(x), f] \nabla +$$

$$\nabla (x), f (x) B + (x) B(x), f \nabla + (x) B(x) f] \quad \rho_{m,1}^{\leftarrow \nabla} =$$

$$\frac{2}{1} [\cancel{(x) B(x) f} - (\nabla (x), B + (x) B) ((x), f \nabla + (x) f)] \quad \rho_{m,1}^{\leftarrow \nabla} =$$

$$\frac{\cancel{(x) B(x) f} - \nabla \frac{(x) B(x) f - (\nabla + x) B(\nabla + x) f}{\nabla}}{(x) f - (\nabla + x) f} \quad \rho_{m,1}^{\leftarrow \nabla} = \frac{x \rho}{(x) \pm \rho}$$

$$(x) B(x) f = (x) f$$